① Aspect 5 (Highest Relevance)

Interpret mathematical outcomes in a variety of formats; compare or evaluate representations

The core task of this question is:

> *"Which one of the following graphs represents the new memory stick's disk status?"*

Students must:

- Interpret disk usage data presented in **table form** (for the old memory stick)
- **Translate and map** this data into a **graphical representation** for the new memory stick
- **Compare and evaluate** multiple graphical options (A, B, C, and D)

This is a典型的 *(typical)* **multiple-representation interpretation task**, making Aspect 5 the most central aspect assessed.

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② Aspect 9

Understand the relationship between context and representation

The problem context involves:

- Deleting and adding files (dynamic changes in stored content)
- Transferring all content to a **larger, empty 2 GB memory stick**
- Representing disk usage as "used vs free space"

Students must understand that:

- The **same set of files** will result in **different proportional representations** on storage devices of different capacities
- The proportions shown in the graphs must be consistent with the real-world storage situation

This reflects a deep understanding of the **context–representation relationship**.

③ Aspect 1

Interpret a mathematical result back into the real-world context

Although the task does not require students to compute a single numerical answer, they must still:

- Interpret the table data as representing actual stored content
- Understand the real meaning of "transferring all files to the new memory stick"
- Decide which graph best matches the **real-world disk status** of the new device

Thus, mathematical information is interpreted through the **real-world storage context**, not treated in isolation.

④ Aspect 2

Identify whether a mathematical result makes sense given the context

When selecting a graph, students must judge reasonableness, for example:

- Eliminating graphs that imply usage exceeding 2 GB
- Rejecting representations with clearly unrealistic proportions
- Considering the condition that the new memory stick was initially empty

This involves evaluating whether a representation is **feasible and contextually valid**.

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⑤ Aspect 7

Construct and communicate explanations and arguments in context

Although the question is multiple-choice, high-quality responses (especially in teaching or assessment contexts) often require students to explain:

- Why other options are incorrect
- Why the chosen graph best represents the situation

This demonstrates the ability to **construct and communicate reasoning** within a contextual framework.

① Aspect 5（相关性最高）

Interpret mathematical outcomes in a variety of formats; compare or evaluate representations

📌 本题的核心任务是：

> *"Which one of the following graphs represents the new memory stick's disk status?"*

学生必须：

- 理解 **表格形式** 中给出的磁盘使用数据（旧 U 盘）
- 将这些数据 **转换并映射** 到 **图像形式**（新 U 盘的磁盘状态图）
- 在 **A、B、C、D 四个图像表示之间进行比较和判断**

👉 这是一个非常典型的 **多重表示比较与选择任务**，直接对应 Aspect 5，是本题最核心的 interpretation 能力。

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② Aspect 9

Understand the relationship between context and representation

📌 题目情境包括：

- 删除和新增文件（动态变化的存储内容）
- 从旧 U 盘转移到一个 **容量更大的新 U 盘（2GB）**
- 磁盘"使用 / 空余"状态的表示

学生必须理解：

- 同样的文件内容，在 **不同容量的存储设备** 上会产生 **不同的比例表示**
- 图像中"已用空间 / 空闲空间"的比例必须与真实情境一致

👉 这体现了对 **情境—表示关系** 的深刻理解。

③ Aspect 1

Interpret a mathematical result back into the real-world context

📌 虽然题目不是直接要求计算一个数值答案，但学生仍需要：

- 将表格中的数据解释为"实际存储了多少内容"
- 理解"全部内容转移到新 U 盘后"的真实含义
- 判断哪一个图像最符合"新 U 盘当前状态"

👉 数学信息的意义必须通过 **真实的存储情境** 来理解，而不是孤立看待数据。

④ Aspect 2

Identify whether a mathematical result makes sense given the context

📌 在选择图像时，学生必须进行合理性判断，例如：

- 是否出现“已用空间超过 2GB”的不合理情况
- 是否图像显示的使用比例明显过大或过小
- 是否忽略了“新 U 盘是空的，后来才转移内容”这一前提

👉 这是对结果 **是否符合现实约束** 的判断。

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⑤ Aspect 7

Construct and communicate explanations and arguments in context

📌 虽然题目本身是选择题，但在教学或高质量作答中，通常要求学生能够解释：

- 为什么排除其他选项
- 为什么所选图像最合理

👉 这体现了在具体情境中 **构建解释和论证的能力**。